TUTORIAL 6

Bases, Invertibility, and Elementary Matrices

MOTIVATION AND GUIDING IDEA

Motivation. A basis turns the construction of a vector into the determination of unique coefficients. An invertible matrix guarantees that every input can be reconstructed from its output. Elementary row operations are reversible steps, so row reduction provides both a reconstruction algorithm and an algebraic explanation of why it works.

NOTATION AND CONVENTIONS

Notation guide. For an ordered basis $B = (b_1, \ldots, b_n)$, the notation $[v]_B = (c_1, \ldots, c_n)^T$ means $v = \sum_{j=1}^{n} c_j b_j$. A^{-1} denotes the inverse, A^T the transpose, and $(A \mid I_n)$ an augmented matrix. $R_i \leftrightarrow R_j$ interchanges rows; $R_i \leftarrow \lambda R_i$ requires $\lambda \neq 0$; $R_i \leftarrow R_i + \lambda R_j$ requires $i \neq j$. rank A is the number of pivots in a row-echelon form. Distinguish a matrix unit E_{ij} from an entire elementary matrix E.

DEFINITION 6.1

Hypotheses and tools. Let V have dimension $n < \infty$ over $\mathbb{F}$. Every independent set in V can be extended to a basis; relative to an ordered basis, every $v \in V$ has a unique expression. If $A \in M_n(\mathbb{F})$, $n \geq 1$, is invertible, then $A^{-1} \in M_n(\mathbb{F})$ satisfies $AA^{-1} = A^{-1}A = I_n$. The inverse is unique: if B, C are inverses, then $B = B(AC) = (BA)C = C$.
The following are equivalent: A is invertible; its reduced row-echelon form is I_n; rank $A = n$; $Ax = b$ has a unique solution for every $b \in \mathbb{F}^n$; $Ax = 0$ has only the zero solution; the columns of A form a basis of $\mathbb{F}^n$. For square matrices of the same order, $(AB)^{-1} = B^{-1}A^{-1}$ when both factors are invertible.
If row operations are applied in the order $E_1, E_2, \ldots, E_k$, the result is $E_k \cdots E_2 E_1 A$. If this equals I_n, then $A^{-1} = E_k \cdots E_1$, whereas $A = E_1^{-1} \cdots E_k^{-1}$. The factor order is part of the assertion.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Develop methods for choosing and extending bases, recognize invertible matrices and calculate their inverses, and connect row reduction with products of elementary matrices. Check each solution through its meaning for a system of equations.

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6.1 Bases and Dimension

DEFINITION 6.2

Let V be a vector space over a field $\mathbb{F}$. A set $\{u_1, \ldots, u_n\} \subseteq V$ is a **basis** of V if it is linearly independent and spans V.

A finite-dimensional vector space can have many bases, but all its bases have the same number of elements. This number is the **dimension** of the space.

EXAMPLE 6.1

$$\dim(\mathbb{F}^n) = n, \qquad \dim M_{m\times n}(\mathbb{F}) = mn, \qquad \dim \mathbb{F}_d[x] = d+1.$$

For example, the standard basis of $\mathbb{R}^4$ is

$$e_1 = (1,0,0,0)^T;\ e_2 = (0,1,0,0)^T;\ e_3 = (0,0,1,0)^T;\ e_4 = (0,0,0,1)^T.$$

The standard basis of $\mathbb{R}_1[x]$ is $(1, x)$.

RECALL AND TOOLS

Let V be a vector space over a field $\mathbb{F}$ with finite dimension, $\dim V < \infty$.

1. If B_1, B_2 are bases of V, then $|B_1| = |B_2|$. Thus dimension is well-defined: $\dim V$ is the cardinality of any basis of V.
2. If $W = \{w_1, \ldots, w_m\} \subseteq V$ is linearly independent, then
$$|W| = m \leq \dim V.$$
One cannot choose more independent vectors than the number of vectors in a basis.
3. If $W = \{w_1, \ldots, w_m\} \subseteq V$ spans V, that is, $V = \operatorname{span}(W)$, then
$$|W| = m \geq \dim V.$$
Every spanning set therefore has at least $\dim V$ vectors.
4. Suppose $W = \{w_1, \ldots, w_m\} \subseteq V$ satisfies:
   - $\dim V = |W| = m$;
   - W is either linearly independent or spans V.

   Then W is a basis of V. If W is independent and has exactly $\dim V$ vectors, no further vector can be added while preserving independence, so W already spans. If W spans and has exactly $\dim V$ vectors, there can be no nontrivial zero linear combination of its elements, so they are independent.

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PROPOSITION 6.1

An additional important fact is the following. Let V have dimension n, and let $W = \{w_1, \ldots, w_k\} \subseteq V$ be linearly independent, with $k \leq n$. There is a set

$$U = \{u_{k+1}, \ldots, u_n\} \subseteq V$$

such that

$$B = W \cup U = \{w_1, \ldots, w_k, u_{k+1}, \ldots, u_n\}$$

is a basis of V. Thus **every linearly independent set in a finite-dimensional space can be extended to a basis**. This is a central tool for constructing bases and calculating dimensions.

RECALL AND TOOLS

To establish that a set is a basis of a space of known finite dimension n, it suffices to verify any two of:

1. Linear independence.
2. Spanning the space.
3. Having exactly n elements, the dimension of the space.

EXERCISE 6.1

Please determine whether these vectors form a basis of $\mathbb{C}^3$. If not, please give a dependence relation.

$$v_1 = (1, 0, 2i)^T, \quad v_2 = (1 - i, 1, 0)^T, \quad v_3 = (5 + 2i, 1 + 2i, -2 + 4i)^T.$$

WORKED SOLUTION

The first two vectors are independent: in $av_1 + bv_2 = 0$, the second coordinate gives $b = 0$, and then the first gives $a = 0$. Computing coordinates gives

$$(2 + i)v_1 + (1 + 2i)v_2 = v_3.$$

$$(2 + i) + (1 + 2i)(1 - i) = 5 + 2i, \qquad (2 + i)2i = -2 + 4i.$$

Thus $(2 + i)v_1 + (1 + 2i)v_2 - v_3 = 0$ is nontrivial. The rank is 2, so the triple is not a basis of $\mathbb{C}^3$.

EXERCISE 6.2

Please determine whether $B = (x^2 + 2x,\ x - 3,\ 2)$ is a basis of $\mathbb{R}_2[x]$ and of $\mathbb{R}_3[x]$. In the first case, please find the coordinates of $p(x) = ax^2 + bx + c$.

WORKED SOLUTION

Write p as a combination with coefficients α, β, γ and compare coefficients:

$$ax^2 + bx + c = \alpha(x^2 + 2x) + \beta(x - 3) + 2\gamma.$$

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EXERCISE 6.2 CONTINUED

$$\alpha = a, \quad 2\alpha + \beta = b, \quad -3\beta + 2\gamma = c.$$

$$[p]_B = (a,\ b - 2a,\ (c + 3b - 6a)/2)^T.$$

For each a, b, c there is a unique coordinate triple, so B is a basis of $\mathbb{R}_2[x]$. Every combination has degree at most 2 and cannot equal x^3, so this is not a basis of $\mathbb{R}_3[x]$.

EXERCISE 6.3

Work in $M_3(\mathbb{R})$. The matrix unit E_{ij} has entry 1 at (i, j) and zero elsewhere. Define

$$D_i = E_{ii} \quad (1 \le i \le 3), \qquad S_{ij} = E_{ij} + E_{ji}, \quad K_{ij} = E_{ij} - E_{ji} \quad (1 \le i < j \le 3).$$

$$\mathcal{B}_s = \{D_1, D_2, D_3, S_{12}, S_{13}, S_{23}\}, \qquad \mathcal{B}_a = \{K_{12}, K_{13}, K_{23}\}.$$

Please prove that $\mathcal{B}_s$ is a basis of the symmetric matrices and $\mathcal{B}_a$ is a basis of the skew-symmetric matrices. Please prove that their union is a basis of $M_3(\mathbb{R})$ and please compute the three dimensions.

WORKED SOLUTION

We have $D_i^T = D_i$, $S_{ij}^T = S_{ij}$, and $K_{ij}^T = -K_{ij}$. Every symmetric $A = (a_{ij})$ has the representation

$$A = \sum_{i=1}^{3} a_{ii} D_i + \sum_{1 \le i < j \le 3} a_{ij} S_{ij}.$$

The diagonal and upper-diagonal entries determine all coefficients uniquely. Thus the family spans and is independent, giving a basis of size $3 + 3 = 6$.
For skew-symmetric A, the diagonal is zero and $a_{ji} = -a_{ij}$. Hence

$$A = \sum_{1 \le i < j \le 3} a_{ij} K_{ij}.$$

Again, the entries above the diagonal uniquely determine the coefficients, so $\mathcal{B}_a$ is a basis of size 3.
Every $A \in M_3(\mathbb{R})$ satisfies

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$

The first summand is symmetric and the second skew-symmetric, so the union spans all matrices. The intersection of the two subspaces is $\{0\}$: $A^T = A = -A$ gives $2A = 0$, hence $A = 0$. Any zero combination of the union therefore splits into a zero combination in each basis, forcing every coefficient to vanish. The union is a basis, with dimensions

$$\dim \mathrm{Sym}_3(\mathbb{R}) = 6, \quad \dim \mathrm{ASym}_3(\mathbb{R}) = 3, \quad \dim M_3(\mathbb{R}) = 9 = 6 + 3.$$

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6.2 The Dimension Formula

PROPOSITION 6.2

Let V be a finitely generated vector space over a field $\mathbb{F}$, and let U be a subspace of V.

1. U is finitely generated and $\dim U \leq \dim V$.
2. $\dim U = \dim V$ if and only if $U = V$.

The dimension formula:

PROPOSITION 6.3

If V is finite-dimensional over $\mathbb{F}$ and $U, W \subseteq V$ are subspaces, then

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

EXERCISE 6.4

1. Let V have dimension n, and let U, W be subspaces with $\dim U = n - 1$ and $W \nsubseteq U$. Please prove $V = U + W$ and $\dim(U \cap W) = \dim W - 1$.
2. $V = \mathbb{R}_2[X]$, $U = \operatorname{span}\{X, X^2 + 9\}$, $W = \{p \in \mathbb{R}_2[X] : p'(-3) = 0\}$. Please compute the dimensions of the sum and intersection, and please determine whether the sum equals V and is direct.

WORKED SOLUTION

1. Choose $w \in W \setminus U$. Then $U \subsetneq U + W \subseteq V$, so

$$n - 1 = \dim U < \dim(U + W) \leq n.$$

Thus $\dim(U + W) = n$ and $U + W = V$. The dimension formula gives

$$n = (n - 1) + \dim W - \dim(U \cap W),$$

hence $\dim(U \cap W) = \dim W - 1$.

2. The generators of U are independent, so $\dim U = 2$. Also $1 \in W \setminus U$. For W,

$$p(X) = aX^2 + bX + c \in W \iff -6a + b = 0 \iff b = 6a.$$

$$W = \operatorname{span}\{1, X^2 + 6X\}, \qquad \dim W = 2.$$

Part 1 gives $U + W = V$. The sum is not direct, since

$$(X + 3)^2 = (X^2 + 9) + 6X \in U, \qquad \frac{d}{dX}(X + 3)^2\Big|_{X=-3} = 0.$$

$$U \cap W = \operatorname{span}\{(X + 3)^2\}, \quad \dim(U \cap W) = 1, \quad \dim(U + W) = 2 + 2 - 1 = 3.$$

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MOTIVATION AND GUIDING IDEA

From a basis to unique reconstruction. The columns of a square matrix form a basis if and only if every right-hand-side vector can be expressed uniquely using those columns. This is the meaning of invertibility and explains why an inverse solves a system for every right-hand side.

6.3 Invertible Matrices

DEFINITION 6.3

A matrix $A \in M_n(\mathbb{F})$, with $n \geq 1$, is **invertible** if there is $B \in M_n(\mathbb{F})$ satisfying $AB = BA = I_n$.

The unique inverse B is denoted A^{-1}.

Basic Properties

RECALL AND TOOLS

- If A is invertible, so is A^{-1}.
- If A, B are invertible, so is AB, and $(AB)^{-1} = B^{-1}A^{-1}$.
- If A is invertible, so is A^T, and $(A^T)^{-1} = (A^{-1})^T$.
- The identity is invertible, with $I^{-1} = I$.
- A square zero matrix of positive order is not invertible.

6.3.1 Exercises on the Algebra of Invertible Matrices

EXERCISE 6.5

Let A, B be square matrices, with A invertible, such that $A = B^2$ and $B = A^2$. Please prove that B is the inverse of A, that is, $B = A^{-1}$.

WORKED SOLUTION

We have

$$\begin{aligned} AB &= AA^2 = A^3, \\ A^3 &= A^3(AA^{-1}) = A^4A^{-1} \\ &= (A^2A^2)A^{-1} = B^2A^{-1} = AA^{-1} = I. \end{aligned}$$

Since $B = A^2$, also $BA = A^2A = A^3 = AB = I$. Therefore $AB = BA = I$, and $B = A^{-1}$.

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EXERCISE 6.6

Let A, B be square matrices such that AB is invertible. Please prove that both A and B are invertible. **This is the converse of the product assertion from the lecture.**

WORKED SOLUTION

Use the fact that for square matrices of the same order, a one-sided inverse is also a two-sided inverse. Indeed, $AX = I$ implies $n = \operatorname{rank}(AX) \leq \operatorname{rank}(A) \leq n$, so A is invertible. From $(AB)(AB)^{-1} = I$, obtain $A[B(AB)^{-1}] = I$. Thus A is invertible, with inverse $B(AB)^{-1}$. Similarly, $(AB)^{-1}(AB) = I$ gives $[(AB)^{-1}A]B = I$. Thus B is invertible, with inverse $(AB)^{-1}A$.

6.3.2 An Algorithm for Finding the Inverse

PROPOSITION 6.4

A square matrix A is invertible if and only if it is row-equivalent to the identity matrix.

Algorithm

1. Form $(A \mid I)$ and row-reduce, aiming to put the left block into **reduced row-echelon form**.
2. Inspect the resulting left block:
   - If it is I, then A is invertible, and the right block is its inverse: $(I \mid A^{-1})$.
   - If it is not I, then A is not invertible.

EXERCISE 6.7

Please determine which matrices are invertible. For each invertible matrix, please find its inverse by row reduction and please verify the result.

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 6 \\ -2 & -4 \end{pmatrix}.$$

WORKED SOLUTION

$$\left(\begin{array}{ccc|ccc} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right).$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 + 2R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & 1 & 2 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right).$$

$$A^{-1} = \begin{pmatrix} -2 & 1 & 2 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \qquad AA^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2+2 & 1 & 2-2 \\ 1-1 & 0 & 1 \end{pmatrix} = I_3.$$

In B, the second row is $-2/3$ times the first. The operation $R_2 \leftarrow R_2 + (2/3)R_1$ creates a zero row, so B is not invertible.

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6.4 Elementary Matrices

DEFINITION 6.4

- Elementary matrices are square matrices representing the elementary row operations used in row reduction.
- Applying an elementary row operation to a matrix is equivalent to multiplying it **on the left** by the corresponding elementary matrix.
- Each elementary matrix is obtained by applying its row operation to the identity matrix.

Consequence. Matrices A, B are row-equivalent if and only if there is a finite sequence of elementary matrices such that $B = E_1 E_2 \cdots E_k A$.

1. **Interchanging rows** i and j is represented by $E^{(1)}_{i,j}$:

$$E^{(1)}_{i,j} = \begin{pmatrix} 1 & & & & \\ & 0 & & 1 & \\ & & 1 & & \\ & 1 & & 0 & \\ & & & & 1 \end{pmatrix}.$$

   - This matrix is its own inverse: interchange the same two rows again.

2. **Multiplying row** i by a nonzero scalar λ is represented by $E^{(2)}_{i,\lambda}$:

$$E^{(2)}_{i,\lambda} = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & \lambda & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}.$$

   - Its inverse is $E^{(2)}_{i,\lambda^{-1}}$: multiply the same row by the reciprocal scalar.

3. **Adding** λ **times row** j to row i, with $i \neq j$, is represented by $E^{(3)}_{i,j,\lambda}$. The entry λ is in row i, column j:

$$E^{(3)}_{i,j,\lambda} = \begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & \lambda & \\ & & & \ddots & & \\ & & & & 1 & \\ & & & & & 1 \end{pmatrix}.$$

   - Its inverse is $E^{(3)}_{i,j,-\lambda}$: add $-\lambda$ times row j to row i.

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EXERCISE 6.8

Please use the preceding reduction to write $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}$ as a product of elementary matrices.

WORKED SOLUTION

The reduction operations are $R_1 \leftrightarrow R_2$, then $R_3 \leftarrow R_3 - R_2$, then $R_1 \leftarrow R_1 + 2R_3$. Their matrices are

$$E_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$E_3E_2E_1A = I_3, \qquad A = E_1^{-1}E_2^{-1}E_3^{-1}$$

$$= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The last two factors multiply to $\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$. The first factor swaps the first two rows and recovers A. Each factor is elementary.

EXERCISE 6.9

Let $A, B, C \in \mathbb{R}^{n\times n}$ be invertible, and let $\alpha \in \mathbb{R}$ with $\alpha \neq 0$. Define

$$D := \alpha A^3 B^T C^2 A.$$

1. Please prove that D is invertible.
2. Please find an explicit expression for D^{-1} in terms of A, B, C, α.

WORKED SOLUTION

1. Each of $\alpha I_n, A, A^3, B^T, C^2$ is invertible:
   - $\alpha \neq 0$, so αI_n is invertible.
   - A is invertible by assumption, so A^3 is invertible.
   - Since B is invertible, B^T is invertible with $(B^T)^{-1} = (B^{-1})^T$.
   - Since C is invertible, C^2 is invertible with $(C^2)^{-1} = (C^{-1})^2$.

   A product of invertible matrices is invertible, so D is invertible.
2. Write the ordered product $D = \alpha A^3 B^T C^2 A$. The inverse of a product is the product of the inverses in reverse order:

$$(XYZ)^{-1} = Z^{-1}Y^{-1}X^{-1}.$$

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EXERCISE 6.9 CONTINUED

Thus

$$D^{-1} = A^{-1}(C^2)^{-1}(B^T)^{-1}(A^3)^{-1}(\alpha I_n)^{-1}.$$

Compute each factor:

$$(C^2)^{-1} = (C^{-1})^2, \quad (B^T)^{-1} = (B^{-1})^T, \quad (A^3)^{-1} = (A^{-1})^3, \quad (\alpha I_n)^{-1} = \frac{1}{\alpha} I_n.$$

Substitution gives

$$D^{-1} = A^{-1}(C^{-1})^2(B^{-1})^T(A^{-1})^3 \cdot \frac{1}{\alpha} I_n = \frac{1}{\alpha} A^{-1}(C^{-1})^2(B^{-1})^T(A^{-1})^3.$$

This is the required expression for D^{-1}.

EXERCISE 6.10

Let

$$A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \in \mathbb{R}^{n \times n}.$$

Please prove that A is invertible if and only if every $\lambda_i \neq 0$, and please find an explicit formula for A^{-1}.

WORKED SOLUTION

If A is invertible, then every $\lambda_i \neq 0$. Suppose A is invertible and, for a contradiction, that $\lambda_k = 0$ for some index k. Consider the standard vector

$$\vec{e}_k = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

with 1 in position k and zero in every other coordinate. Then

$$A\vec{e}_k = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \lambda_k \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \vec{0},$$

since $\lambda_k = 0$. Thus $A\vec{e}_k = \vec{0}$ with $\vec{e}_k \neq \vec{0}$, a nontrivial solution of $Ax = 0$. If A were invertible, left multiplication by A^{-1} would give $e_k = A^{-1}0 = 0$, a contradiction. Therefore every $\lambda_i \neq 0$.

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EXERCISE 6.10 CONTINUED

If every $\lambda_i \neq 0$, then A is invertible; we find its inverse. Assume all these scalars are nonzero. Define

$$B := \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right) = \begin{pmatrix} \frac{1}{\lambda_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\lambda_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\lambda_n} \end{pmatrix}.$$

Calculate

$$\begin{aligned} AB &= \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right) \\ &= \operatorname{diag}\left(\lambda_1 \frac{1}{\lambda_1}, \ldots, \lambda_n \frac{1}{\lambda_n}\right) \\ &= \operatorname{diag}(1, \ldots, 1) = I_n. \end{aligned}$$

The same calculation in reverse order gives $BA = I_n$. Hence B is the inverse of A, so

$$A^{-1} = B = \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right).$$

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6.4.1 Equivalent Conditions

PROPOSITION 6.5

For a square matrix $A \in M_n(\mathbb{F})$, the following conditions are equivalent:

1. A is invertible.
2. A is row-equivalent to I_n.
3. $\operatorname{rank}(A) = n$.
4. For every column vector $\vec{b} \in \mathbb{F}^n$, the system $A\vec{x} = \vec{b}$ has a solution.
5. For every column vector $\vec{b} \in \mathbb{F}^n$, the system $A\vec{x} = \vec{b}$ has a unique solution $\vec{x} \in \mathbb{F}^n$.

6.4.2 The Connection with the Homogeneous System

In the preceding assertions, take $\vec{b} = \vec{0}$. There is then a unique solution, the trivial solution. Thus:

PROPOSITION 6.6

Another equivalent condition: $A \in M_n(\mathbb{R})$ is invertible if and only if its homogeneous system has only the trivial solution. In other words, the only solution of $A\vec{x} = \vec{0}$ is $\vec{x} = \vec{0}$.

EXERCISE 6.11

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}, \qquad b = (\alpha, 2\gamma, 3\phi)^T, \qquad \alpha, \gamma, \phi \in \mathbb{R}.$$

Please solve $Ax = b$ in terms of the parameters and please prove that the system is homogeneous if and only if its unique solution is zero.

WORKED SOLUTION

$$x = A^{-1}b = \begin{pmatrix} -2 & 1 & 2 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ 2\gamma \\ 3\phi \end{pmatrix} = \begin{pmatrix} -2\alpha + 2\gamma + 6\phi \\ \alpha \\ -\alpha + 3\phi \end{pmatrix}.$$

Indeed, $x_2 = \alpha$, $x_1 - 2x_3 = 2\gamma$, and $x_2 + x_3 = 3\phi$. The solution is unique because A is invertible. The system is homogeneous exactly when $b = 0$, equivalently $\alpha = \gamma = \phi = 0$, which gives $x = 0$. Conversely, if $x = 0$ is a solution, then $b = A0 = 0$.

EXERCISE 6.12

$$A = I_3 + N, \qquad N = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}.$$

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EXERCISE 6.12 CONTINUED

Please compute N^2, N^3, please find A^{-1} without row reduction, and please solve $Ax = (3,8,3)^T$.

WORKED SOLUTION

$$N^2 = \begin{pmatrix} 0 & 0 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = 0.$$

$$(I+N)(I-N+N^2) = I - N + N^2 + N - N^2 + N^3 = I,$$

$$(I-N+N^2)(I+N) = I + N - N - N^2 + N^2 + N^3 = I.$$

$$A^{-1} = I - N + N^2 = \begin{pmatrix} 1 & 1 & -3 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$x = A^{-1}\begin{pmatrix} 3 \\ 8 \\ 3 \end{pmatrix} = \begin{pmatrix} 3+8-3\cdot 3 \\ 8-3\cdot 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}.$$

The only nonzero entry of N^2 is $(-1)\cdot 3 = -3$. Verification gives $Ax = (2-(-1), -1+3\cdot 3, 3)^T = (3,8,3)^T$.

Additional practice

EXERCISE 6.13 FOUNDATION

Let $w_1 = (1,2,0)^T$, $w_2 = (0,1,-1)^T$, $e_3 = (0,0,1)^T$. Please prove that $B = (w_1, w_2, e_3)$ is a basis of $\mathbb{R}^3$. Please find the inverse of its column matrix by row reduction and please compute $[b]_B$ for $b = (3,8,5)^T$.

WORKED SOLUTION

The equation $aw_1 + bw_2 + ce_3 = 0$ gives $a = 0$, $2a + b = 0$, and $-b + c = 0$, so $a = b = c = 0$. Three independent vectors in $\mathbb{R}^3$ form a basis.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}.$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 & 1 \end{array}\right).$$

$$A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -2 & 1 & 1 \end{pmatrix}, \qquad [b]_B = A^{-1}\begin{pmatrix} 3 \\ 8 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ -6+8 \\ -6+8+5 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 7 \end{pmatrix}.$$

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EXERCISE 6.13 FOUNDATION CONTINUED

Indeed, $3w_1 + 2w_2 + 7e_3 = (3, 6+2, -2+7)^T = (3, 8, 5)^T$. The representation is unique because B is a basis.

EXERCISE 6.14 CHALLENGE

Let $u, v \in \mathbb{R}^n$, $n \geq 1$, and define $A = I_n + uv^T$.

1. Please prove that A is invertible if and only if $1 + v^T u \neq 0$, and please find its inverse in this case without using a determinant.
2. Please apply the formula to $u = (1, -1, 0)^T$, $v = (2, 0, 1)^T$, and please solve $Ax = (5, 0, -1)^T$.

WORKED SOLUTION

Write $P = uv^T$ and $c = v^T u \in \mathbb{R}$. Associativity gives

$$P^2 = (uv^T)(uv^T) = u(v^T u)v^T = cP.$$

If $1 + c \neq 0$, define $B = I_n - P/(1+c)$. Then

$$AB = (I_n + P)\left(I_n - \frac{P}{1+c}\right) = I_n + \left(1 - \frac{1}{1+c} - \frac{c}{1+c}\right)P = I_n.$$

Multiplication in the opposite order gives exactly the same expansion, so $BA = I_n$. Therefore

$$A^{-1} = I_n - \frac{uv^T}{1 + v^T u}.$$

If $1 + c = 0$, then $c = -1$, so $u \neq 0$, whereas

$$Au = u + u(v^T u) = (1 + c)u = 0.$$

An inverse would imply $u = 0$, a contradiction. This proves both directions.
For the given example,

$$v^T u = 2, \quad A = \begin{pmatrix} 3 & 0 & 1 \\ -2 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ \frac{2}{3} & 1 & \frac{1}{3} \\ 0 & 0 & 1 \end{pmatrix}.$$

$$x = A^{-1}\begin{pmatrix} 5 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} (5+1)/3 \\ 10/3 - 1/3 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}.$$

Verification: $Ax = (3 \cdot 2 - 1, -2 \cdot 2 + 3 + 1, -1)^T = (5, 0, -1)^T$.

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EXERCISE 6.15 CHALLENGE

Let $A \in M_n(\mathbb{R})$ satisfy $A^m = I_n$ for an integer $m \geq 2$, and define

$$P = \frac{1}{m} \sum_{k=0}^{m-1} A^k.$$

Please prove that $AP = PA = P$ and $P^2 = P$. Please find the inverse of $I_n + P$.

WORKED SOLUTION

Multiplication by A shifts the powers, so

$$AP = \frac{1}{m}(A + A^2 + \cdots + A^m) = P.$$

Since powers of A commute with A, we also have $PA = P$. Induction gives $A^k P = P$ for every $k \geq 0$. Thus

$$P^2 = \frac{1}{m} \sum_{k=0}^{m-1} A^k P = P.$$

Consequently

$$(I_n + P)(I_n - P/2) = I_n + P/2 - P^2/2 = I_n.$$

The reverse product is also I_n, since both factors are polynomials in P. Therefore $(I_n+P)^{-1} = I_n - P/2$.

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EXERCISE 6.16 CHALLENGE

Let $A, B \in M_n(\mathbb{R})$ and define the block matrix

$$C = \begin{pmatrix} A & B \\ B & A \end{pmatrix}.$$

Please prove that C is invertible if and only if $A+B$ and $A-B$ are invertible. In that case, please find C^{-1} in terms of $(A+B)^{-1}$ and $(A-B)^{-1}$.

WORKED SOLUTION

Define

$$J = \begin{pmatrix} I_n & I_n \\ I_n & -I_n \end{pmatrix}.$$

Since $J^2 = 2I_{2n}$, we have $J^{-1} = J/2$. Block multiplication gives

$$CJ = \begin{pmatrix} A+B & A-B \\ A+B & -(A-B) \end{pmatrix},$$

and therefore

$$J^{-1}CJ = \begin{pmatrix} A+B & 0 \\ 0 & A-B \end{pmatrix}.$$

Multiplication on either side by invertible matrices preserves invertibility. A block diagonal matrix is invertible exactly when both its diagonal blocks are invertible. This proves the equivalence. Writing $X = (A+B)^{-1}$ and $Y = (A-B)^{-1}$, we obtain

$$\begin{aligned} C^{-1} &= J \begin{pmatrix} X & 0 \\ 0 & Y \end{pmatrix} J^{-1} \\ &= \frac{1}{2} \begin{pmatrix} X+Y & X-Y \\ X-Y & X+Y \end{pmatrix}. \end{aligned}$$

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